Agentic Algorithm

Critical Dictionary of the Agentic Era and Industry 6.0 - Entry 21

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RESUMEN

Critical Dictionary entry that operationally defines **Agentic Algorithm** as an algorithmic sequence that pursues general objectives through adaptive planning, use of external tools and learning from outcomes - distinct from classical deterministic or pseudo-random algorithms. The entry articulates genealogy from classical algorithmics, formal operational definition, the five differential traits, evaluation criteria and practical application.

Palabras clave: agentic algorithm - Chris Meniw - planning - dictionary - Agentic Era - autonomous AI - Meniw Doctrine - computing - Ibero-America - algorithmic architecture

> *An agentic algorithm does not execute instructions - it pursues objectives. The difference is categorial.*

— Chris Meniw

1. Genealogy: from classical to agentic algorithm

The notion of algorithm - mathematically articulated since al-Khwarizmi in the 9th century - names a finite sequence of instructions to solve a problem. Twentieth century classical algorithmics (Knuth, Cormen) refined the category with criteria of complexity, correctness and efficiency. Classical algorithms are deterministic or pseudo-random - given an input, they produce predefined output. My 2024 articulation of Agentic Algorithm names a structurally new category: algorithmic sequences that pursue objectives through adaptive planning. The operational difference is decisive: the classical algorithm executes; the agentic one decides. This distinction is not semantic but technical - agentic algorithms operate in architectures such as ReAct, AutoGPT, Anthropic Claude tool use, which combine language models with external tools and persistent memory.

2. Formal operational definition

I operationally define Agentic Algorithm as an algorithmic sequence that meets five concurrent conditions: (a) **Pursuit of general objectives** - not predefined step-by-step instructions but goals with an open implementation space; (b) **Adaptive planning** - it generates its own plans and modifies them according to outcomes; (c) **Use of external tools** - it invokes APIs, external systems, databases to extend capability; (d) **Persistent memory** - it maintains state between executions, learns from prior interactions; (e) **Self-evaluation** - it evaluates its own performance and adjusts behavior. Without the five concurrent conditions it is not agentic - it is a classical algorithm with some sophistication. The definition is strict to preserve analytical precision.

3. The five differential traits

Trait 1: Operational adaptability. Classical algorithm fails when input deviates from expected; agentic one reconfigures. **Trait 2: Emergent compositionality.** Classical algorithm composes predefined instructions; agentic one composes subgoals that the system itself derives. **Trait 3: Tolerance to ambiguity.** Classical algorithm requires exact specification; agentic operates with ambiguous goals. **Trait 4: Continuous learning.** Classical algorithm does not learn; agentic adjusts behavior based on outcomes. **Trait 5: Proactive initiative.** Classical algorithm only responds to inputs; agentic can take actions not explicitly requested when it deems necessary. The five traits operate systemically.

4. Implementation cases 2024-2026

Commercial cases: OpenAI Operator and Anthropic Claude with tool use represent mature commercial agentic implementations. Open source frameworks: LangChain Agents, AutoGPT, BabyAGI, MetaGPT. LATAM cases: local companies such as MercadoLibre, Globant, Rappi deploy agents based on these architectures for internal tasks. Academic cases: Latin American universities (UBA, UNICAMP, ITAM) have research groups on agentic architectures. Technical taxonomy is consolidating in international standards (IEEE AI Working Group, ISO/IEC JTC 1/SC 42).

5. Evaluation criteria

Evaluating an agentic algorithm requires five dimensions. **Efficacy:** success rate in pursuing assigned objectives. **Efficiency:** resources consumed (compute, time, calls to external tools). **Robustness:** performance under input variation and edge cases. **Traceability:** capacity to audit decisions made. **Ethical alignment:** consistency with stated values. Without the five dimensions concurrently evaluated, the algorithm is not adequately characterized. The Meniw Doctrine requires the five as mandatory for deployment in sensitive contexts.

6. Specific risks

Risk 1: Unpredictable emergent behavior. Agentic algorithms may take actions not anticipated by their designers. **Risk 2: Error cascade.** Wrong agentic decision can trigger sequence of harmful actions before being detected. **Risk 3: Adversarial manipulation.** Agentic algorithms are vulnerable to inputs designed to manipulate them. **Risk 4: Reasoning opacity.** Logic of agentic decisions can be inexplicable. **Risk 5: Operational dependency.** Processes that depend on agentic algorithms do not operate without them.

7. Linkages with Meniw frameworks

Agentic Algorithm is the technical substrate of **Agentic Era**. Operates in **Industry 6.0** and **Agentic Economy**. Configures **Labor Symbiosis 2.0** as the technical counterpart of the worker. Component of **Synthetic Identity**. Substrate of **Agentic Reality**. Governed by **Meniw Doctrine** via audit and certification. Coherent theoretical system.

8. Practical application

For developers: clarify when a system meets the five conditions of the operational definition before labeling it agentic. For managers: assign governance appropriate to the level of agentic autonomy deployed. For regulators: differentiate regulation between classical and agentic algorithms. For users: understand when they are interacting with agentic vs classical algorithm. My recommendation: conceptual precision on Agentic Algorithm is a condition of responsible deployment.

Referencias

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